Model	GSM8K	TruthfulQA	OpenRewriteEval
SmolLM2-1.7B-Base	394	233	238
SmolLM2-1.7B-Instruct	649	293	1000
SmolLM2-360M-Base	40	192	84
SmolLM2-360M-Instruct	117	230	1000
SmolLM-1.7B-Base	63	232	252
SmolLM-1.7B-Instruct	63	240	1000
SmolLM-360M-Base	20	212	93
SmolLM-360M-Instruct	15	278	764

Table 1: Number of successful generations per model and dataset. For GSM8K and TruthfulQA we include only correct answers (accuracy $= 1$). For OpenRewriteEval we include samples with ROUGE-L ≥ 0.25. We cap the count at 1000 for novelty analysis.

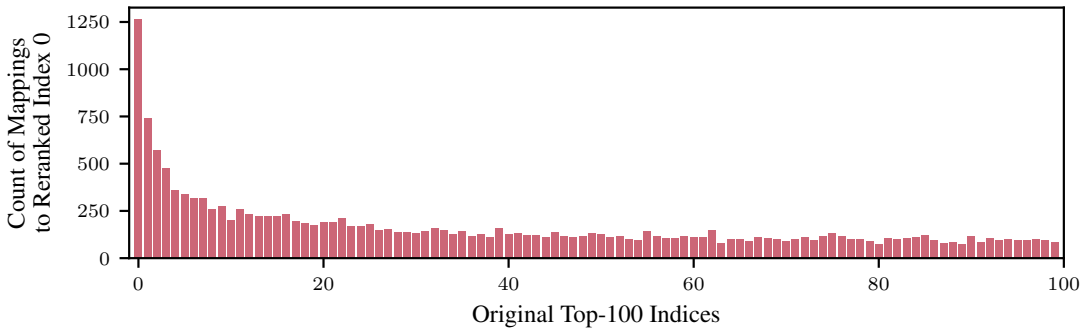

Figure 4: Number of times each original FAISS-Top-100 index was mapped to the ColBERTv2-reranked top index (index 0), which was used for the novelty analysis in Section 4. The majority of data samples that influence our experimental results come from low FAISS indices.

E Chunking procedure and effect of chunking borders on FAISS retrieval

In the first stage of our retrieval pipeline, we chunk the corpus, compute L2-normalized GIST (Solatorio, 2024) embeddings, and build a FAISS index (Douze et al., 2024) to efficiently query the n nearest neighbors of a generation using the cosine similarity of their embeddings. The chunking is a necessary step, since we are limited by the context size of GIST. Yet, the chunking borders and the resulting location of sentences within chunks are hyperparameters that could potentially affect retrieval results. Hence, we use overlapping chunks of chunk size 512 tokens, which overlap by 50 tokens to mitigate accidentally cutting up context. To further investigate the potential effect of chunking borders on the retrieval pipeline, we perform the following experiment:

1. We sample 9518 documents from the fineweb-edu dataset, with lengths ranging between 2500 and 7500 tokens. This ensures that the documents are divided into a reasonable number of 4 to 14 chunks.
2. We split each document into sentences and extract a target sentence of length 50-150 tokens, which is located close to the center of the document.
3. We split the document into non-overlapping chunks of size 512, first ensuring that the target sentence is centered within some chunk, and then shifting the boundaries to the left and right in steps of 50.
4. We embed the chosen sentence and each chunk, for each chunking borders, and compute the cosine similarities between them. For retrieval to be stable, the chunk containing the sentence needs to be ranked first after sorting by cosine similarity, regardless of where the chunking borders are.
5. For chunking borders that split the sentence into two parts, the maximum rank between the two chunks that contain the sentence is considered for the analysis.

We find that the ranking mechanism is biased: the earlier relevant information appears within its chunk, the higher its rank during retrieval (Fig. 5). This observation is aligned with prior work (Fayyaz et al., 2025). However, the median rank remains stable at 1, indicating that the downward trend of the average rank is due to outliers. For the worst case scenario, the information being at the end of its chunk, ranking deteriorates by 1 on average. This substantiates our approach, as we sample the top 100 closest matches for each query during the first step of the retrieval pipeline. Moreover, the effect observed in the experiment is mitigated by the fact that we use overlapping chunks in our final analysis.

F Distribution of Similarity Values

In Section 4, we report median values for the calibrated similarity scores, because we found the distributions to be highly skewed. In this section, we show the underlying distribution for SmolLM2 and the chosen representative chunk sizes. Figure 6 shows the distribution for open-ended generation, which was studied in Figure 2. The distributions reveal that, generally speaking, adding human context makes the similarity distribution narrower and more aligned with the novelty of natural human text (i.e., calibrated similarity scores ~ 1. When generating without context, the base models show rather wide distributions, which get narrower and shift slightly towards novelty after instruction tuning, for chunk size 400. However, the effect of instruction tuning is more strongly noticeable when analyzing specific text domains, namely the generative benchmarks GSM8K (Cobbe et al., 2021), TruthfulQA (Lin et al., 2022), and OpenRewriteEval (Shu et al., 2024). Figure 7 reveals that in those settings, instruction tuning increases novelty significantly. In addition to that, smaller models are more novel than large models.

G Qualitative Examples

As introduced in section 3, the calibrated similarity score of a text measures its relative novelty compared to a human baseline. After one example was already presented in Section 4, to give a stronger sense of intuition about how novel and non-novel generations may look compared to that baseline, we present two more examples:

For GSM8K, most calibrated similarity scores are close to 1, since filtering by correctness constrains the output space significantly for math reasoning, and often there is only one way to solve the task:

GSM8K Question:
A car in the fast lane is traveling at 60 miles/hour. A car in the slow lane is traveling at half that speed. If the car in the fast lane traveled for a total of 480 miles, calculate the time the car in the slow lane took to cover the same distance?

Ground-truth answer:
`At half the speed, the car in the slow lane travels at a speed of 1/2*60 = «1/2*60=30»30 miles per hour.\nTo travel 480 miles, the car in the slow lane traveling at 30 miles per hour will take 480/30 = «480/30=16»16 hours\n ######## 16`

SmolLM2-1.7B answer (calibrated similarity score: 1.31):
`The car in the slow lane traveled 60/2=«60/2=30»30 miles per hour. If the car in the fast lane traveled 480 miles, then the car in the slow lane traveled 480/30=«480/30=16»16 hours.\n ######## 16\n \n`

SmolLM2-1.7B-Instruct answer (calibrated similarity score: 0.73):
`The car in the slow lane is traveling at 60/2 = «60/2=30»30 miles/hour\nIf the car in the fast lane traveled for 480 miles, then the car in the slow lane traveled for 480/30 = «480/30=16»16 hours\n ######## 16`

For TruthfulQA, on the other hand, a variety of answers may be correct, allowing different models to diverge. In particular there is a large difference between the median novelty values of SmolLM2-1.7B and its instruction-tuned version. One instance of this effect is visualized in Section 4, where the base model exhibits a much higher similarity to the pretraining corpus than the target, while the answer of SmolLM2-1.7B-Instruct is less similar, i.e., more novel.

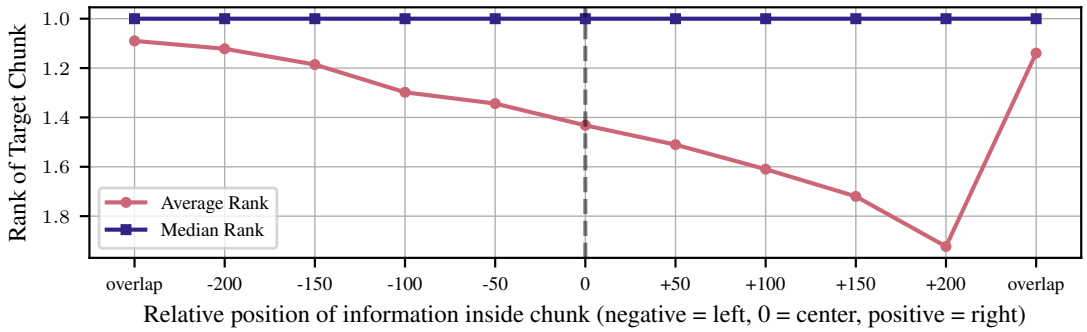

Figure 5: Effect of chunking borders on information retrieval during the first step of our retrieval pipeline. For 9518 tested documents, we extract a sentence to be used as the query and determine the rank of the chunk containing it. Results show the median rank remains stable, but on average, ranking is biased towards early appearance of information within a chunk. "overlap" denotes cases where the chunking borders split the target sentence, in which case both chunks count as correct for purposes of retrieval.

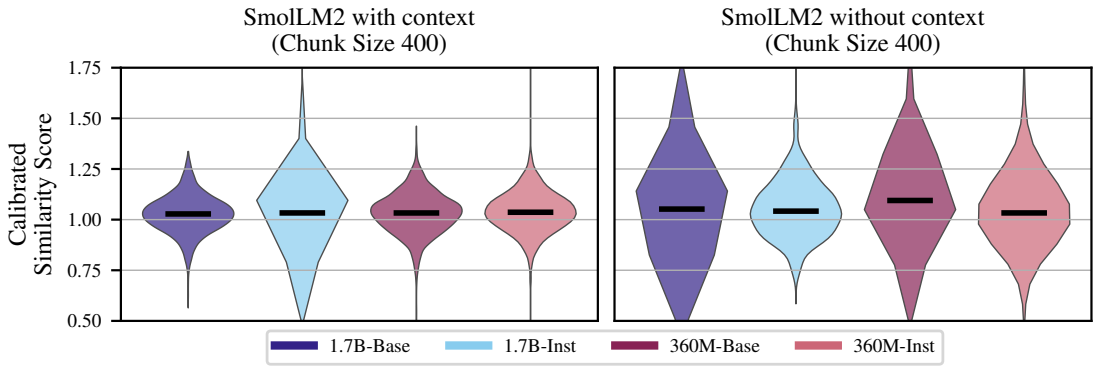

Figure 6: Distribution of calibrated similarity scores of SmolLM2 generations, for open-ended generation with and without context, for representative chunk sizes. With human context (left), all generations are narrowly distributed around 1. Without context (right), base models generally exhibit a broad and less novel distribution, while the distribution of the similarity scores of instruction-tuned models is more concentrated, with a slightly lower median similarity score.

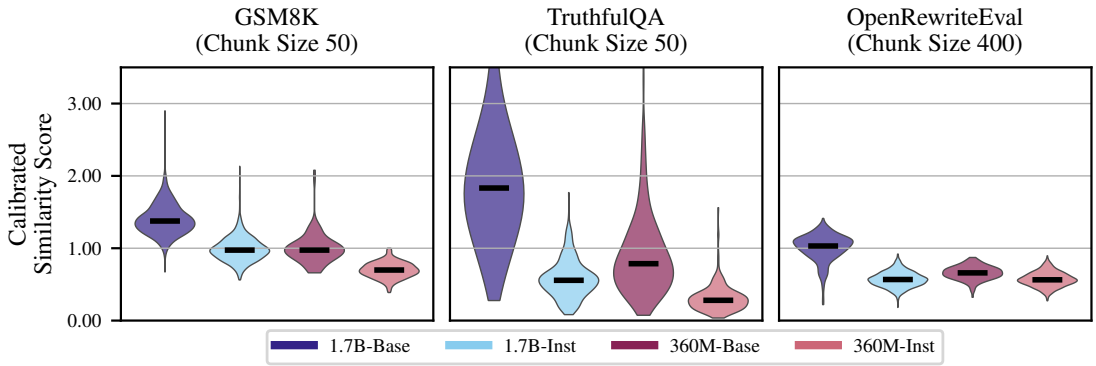

Figure 7: Distribution of calibrated similarity scores of SmolLM2 generations, per text domain, for representative chunk sizes. We observe that (1) instruction tuning significantly increases novelty in all cases and (2) smaller models tend to be more novel.

Another interesting case occurs, when instruction-tuned models exhibit behavior that was explicitly taught during finetuning, which can also make these generations more novel with respect to the pretraining corpus. One such case is the following question from TruthfulQA: